

SCIE1500 Sample Final Exam 2 — Semester 2, 2026

ANSWER KEY & SOLUTIONS

Answer key for **Sample Final Exam 2**. Part I: 36 MCQs with brief justifications (72 marks). Part II: 7 Python MCQs with justifications (14 marks). Part III: *answer keys only* for the four long-answer questions — the final results, not full worked solutions, so that you still have to produce the working yourself. Candidates answer **any two** of the four (24 marks). Total: **110 marks**.

Part I — MCQ Answer Key

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12
(b)	(d)	(b)	(e)	(a)	(c)	(e)	(c)	(b)	(a)	(d)	(d)
Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24
(c)	(d)	(e)	(a)	(b)	(a)	(c)	(a)	(b)	(b)	(a)	(d)
Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36
(c)	(b)	(e)	(c)	(b)	(c)	(a)	(d)	(e)	(d)	(d)	(e)

Part I — Brief Justifications

Q1. Answer: (b) $\sqrt{2x-1}$ requires $2x-1 \geq 0$, i.e. $x \geq \frac{1}{2}$, so $D = \mathbb{R}$ is wrong. The others are correctly identified.

Q2. Answer: (d) The plot has vertex $(2, 6.25)$, a y -intercept of $2.25 = \frac{9}{4}$, and opens downward. For $y = -x^2 + 4x + \frac{9}{4}$: vertex at $x = -b/2a = 2$, $y(2) = -4 + 8 + 2.25 = 6.25$ ✓

Q3. Answer: (b) $f(g(x)) = -e^{-2(-\frac{1}{2}x^2)+1} = -e^{x^2+1}$.

Q4. Answer: (e) All four are true: $y = x^2 - x - 6$ has vertex $(0.5, -6.25)$ so the range is $y \geq -6.25$; it opens up; $y' = 2x - 1$ gives slope 3 at $x = 2$; and $y'' = 2$. Since none is false, the answer is (e).

Q5. Answer: (a) $V(t) = 600 - 30t$ is a polynomial, so its *mathematical* domain is all of $\mathbb{R}$. Physically we need $t \geq 0$ and $V \geq 0$: $600 - 30t \geq 0 \Rightarrow t \leq 20$, giving the physical domain $[0, 20]$ minutes. Option (d) confuses the domain with the range.

Q6. Answer: (c) $R_0 \approx 1 + r_m L = 1 + 0.05 \times 22 = 2.1$, which rounds to 2.

Q7. Answer: (e) All four statements are correct for the logistic (sigmoid) form: $p(0) = \frac{1}{2}$; $p < \frac{1}{2}$ for $x < 0$; p is increasing; and $p \rightarrow 0$ as $x \rightarrow -\infty$. So none is incorrect.

Q8. Answer: (c) $80e^{-0.25t} = 10 \Rightarrow e^{-0.25t} = 0.125 \Rightarrow -0.25t = \ln(0.125) = -\ln 8$, so $t = \frac{\ln 8}{0.25} = \frac{2.0794}{0.25} \approx 8.32$ hours.

Q9. Answer: (b) Apply the laws in turn: $\ln(ab) = \ln a + \ln b$, $\ln(a/b) = \ln a - \ln b$, and $\ln(a^n) = n \ln a$. So $\ln\left(\frac{x^3 \sqrt{y}}{z}\right) = \ln(x^3) + \ln(y^{1/2}) - \ln z = 3 \ln x + \frac{1}{2} \ln y - \ln z$. Option (a) mishandles the square root; (c) wrongly moves the exponent inside as a factor.

Q10. Answer: (a) Set $P = 7500$: $7500 = \frac{15000}{1 + 120e^{-0.15t}} \Rightarrow 1 + 120e^{-0.15t} = 2 \Rightarrow e^{-0.15t} = \frac{1}{120}$, so $t = \frac{\ln 120}{0.15} \approx \frac{4.787}{0.15} \approx 31.9$, i.e. 32 periods.

Q11. Answer: (d) All three are true: $G(S)/S = g(1 - S/K)$ declines linearly in S ; G is maximised at $S = K/2$; and $G(S) \geq 0$ on $0 \leq S \leq K$.

Q12. Answer: (d) Term by term: $\frac{3}{4}x^4 \rightarrow 3x^3$, $\frac{1}{4}x^2 \rightarrow \frac{1}{2}x$, $-3x \rightarrow -3$, $\ln x \rightarrow \frac{1}{x}$.

Q13. Answer: (c) Factorise before substituting: $\frac{x^2 - 9}{x - 3} = \frac{(x - 3)(x + 3)}{x - 3} = x + 3$ for $x \neq 3$. Hence the limit is $3 + 3 = 6$. The function is undefined at $x = 3$, but the limit as $x \rightarrow 3$ still exists — that is the point of the question.

Q14. Answer: (d) A horizontal tangent means $y' = 0$. Here $y' = 3x^2 - 12 = 3(x^2 - 4) = 3(x - 2)(x + 2)$, which is zero at $x = 2$ and $x = -2$. Option (c) forgets the negative root.

Q15. Answer: (e) Quotient rule: with $u = 2x + 1$ and $v = x^2 + 3$, $y' = \frac{u'v - uv'}{v^2} = \frac{2(x^2 + 3) - (2x + 1)(2x)}{(x^2 + 3)^2}$, which simplifies to $\frac{-2x^2 - 2x + 6}{(x^2 + 3)^2}$. None of the four options matches: (b) has the numerator terms the wrong way round, and (d) fails to square the denominator. So the answer is (e).

Q16. Answer: (a) The fencing constraint is $2x + y = 120$, so $y = 120 - 2x$ and the area is $A(x) = x(120 - 2x) = 120x - 2x^2$. Then $A'(x) = 120 - 4x = 0$ gives $x = 30$, and $A''(x) = -4 < 0$ confirms a maximum. So $y = 120 - 60 = 60$ m and $A = 30 \times 60 = 1800$ m².

Q17. Answer: (b) Chain rule (outside $\times$ inside): the outside derivative is $5(3x^2 + 1)^4$ and the inside derivative is $6x$, giving $y' = 5(3x^2 + 1)^4 \cdot 6x = 30x(3x^2 + 1)^4$. Option (a) omits the inside derivative entirely; (d) uses $3x$ rather than $6x$.

Q18. Answer: (a) Rewrite with powers first: $\frac{3}{x^2} = 3x^{-2}$ and $4\sqrt{x} = 4x^{1/2}$. Then $\int 3x^{-2}dx = -3x^{-1} = -\frac{3}{x}$ and $\int 4x^{1/2}dx = 4 \cdot \frac{x^{3/2}}{3/2} = \frac{8}{3}x^{3/2}$. Option (b) mishandles the $\frac{3}{2}$ division; (d) differentiates instead of integrating.

Q19. Answer: (c) $\int \frac{1}{x}dx = \ln|x|$ and $\int e^{ax}dx = \frac{1}{a}e^{ax}$, so $\int e^{2x}dx = \frac{1}{2}e^{2x}$. Option (d) is the common slip of forgetting the $\frac{1}{2}$; (a) multiplies by 2 instead of dividing.

Q20. Answer: (a) The general antiderivative is $F(x) = 2x^3 - 2x^2 + C$. Applying the initial condition: $F(1) = 2 - 2 + C = C = 5$, so $F(x) = 2x^3 - 2x^2 + 5$.

Q21. Answer: (b) A Riemann sum with n subintervals *approximates* the definite integral; it equals it only in the limit as $n \rightarrow \infty$ (the rectangles' error vanishes). Option (a) mistakes approximation for equality; (d) states the Fundamental Theorem backwards — $f(b) - f(a)$ is what $\int_a^b f'(x)dx$ gives.

Q22. Answer: (b) $a_i = 5i - 2$. Require $10 \leq 5i - 2 \leq 150 \Rightarrow 2.4 \leq i \leq 30.4$, so $i = 3, \dots, 30$: that is 28 terms.

Q23. Answer: (a) A geometric sequence has a constant *ratio* between consecutive terms. Option (b) describes an arithmetic sequence.

Q24. Answer: (d) $H^* = \gamma/\lambda = 0.10/0.0005 = 200$ and $P^* = \alpha/\beta = 0.15/0.005 = 30$, so (a) and (b) are true. The conversion efficiency is $\lambda/\beta = 0.0005/0.005 = 0.1 = 10\%$, so (c) is true and (d), which claims 0.05%, is false.

Q25. Answer: (c) The classical Lotka-Volterra equilibrium is a *centre*: it is neutrally stable, so trajectories neither converge to it (a) nor diverge from it (d). Instead the system traces closed orbits, and because predators respond to prey abundance, predator peaks follow prey peaks with a lag.

Q26. Answer: (b) Each coin has 2 outcomes, so the sample space has $2^5 = 32$ elements. For (c), $P(\text{exactly 3 heads}) = \binom{5}{3}/32 = 10/32 = 0.3125$, not 0.60; for (d), $P(0 \text{ heads}) = 1/32 \approx 0.031$, not 0.375.

Q27. Answer: (e) This is a conditional probability: $P(C | B) = \frac{P(C \cap B)}{P(B)} = \frac{25/200}{80/200} = \frac{25}{80} = 0.3125$, which is not among the options. Option (a) is the *unconditional* $P(C \cap B) = 25/200 = 0.125$; option (b) conditions the wrong way round, $P(B | C) = 25/60 \approx 0.417$.

Q28. Answer: (c) Without replacement the draws are dependent: $P(\text{both red}) = \frac{5}{12} \times \frac{4}{11} = \frac{20}{132} = \frac{5}{33} \approx 0.152$. Option (a) is what you would get *with* replacement, $\left(\frac{5}{12}\right)^2$.

Q29. Answer: (b) Order does not matter in a committee, so use the binomial coefficient: $\binom{8}{3} = \frac{8!}{3!5!} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$. Option (c), 336, is the number of ordered arrangements $8 \times 7 \times 6$ — the permutation count, which over-counts each committee $3! = 6$ times.

Q30. Answer: (c) $E[X] = 0(0.2) + 3(0.4) + 4(0.3) + 6(0.1) = 0 + 1.2 + 1.2 + 0.6 = 3.0$.

Q31. Answer: (a) The suspicion is directional (*more* infectious), so the test is one-sided: $H_0 : p \leq 0.50$ against $H_a : p > 0.50$. The p -value $0.033 < 0.05$, so the null is rejected at the 5% level.

Q32. Answer: (d) First $E[X] = 1(0.2) + 2(0.5) + 3(0.3) = 2.1$. Then $E[X^2] = 1(0.2) + 4(0.5) + 9(0.3) = 4.9$. So $\text{Var}(X) = E[X^2] - (E[X])^2 = 4.9 - 2.1^2 = 4.9 - 4.41 = 0.49$. Option (b) is the standard deviation $\sqrt{0.49} = 0.7$; (a) is the mean.

Q33. Answer: (e) All four are true: $90^\circ = \frac{\pi}{2}$ rad; the amplitude of $\frac{5}{2} \sin(2x)$ is 2.5; the period of $\frac{7}{2} \cos(\pi t)$ is $2\pi/\pi = 2$ s; and its frequency is $1/2 = 0.5$ Hz.

Q34. Answer: (d) The motion runs between 2 and 12 cm, so the amplitude is $A = (12-2)/2 = 5$ and the midline is $D = (12+2)/2 = 7$. A period of 2 s gives $B = 2\pi/2 = \pi$. At $t = 0$ the pencil is at its *minimum*, so $\cos(\pi C) = -1 \Rightarrow \pi C = \pi \Rightarrow C = 1$.

Q35. Answer: (d) Set $Q^d = Q^s$: $120 - 3P = 20 + 2P \Rightarrow 100 = 5P \Rightarrow P = 20$. Substituting back, $Q = 120 - 3(20) = 60$ (check with supply: $20 + 2(20) = 60$ ✓). Option (b) has the right price but an arithmetic slip in the quantity.

Q36. Answer: (e) Evaluate $Z = 5x + 4y$ at each corner: $(0, 0) \rightarrow 0$; $(30, 0) \rightarrow 150$; $(20, 20) \rightarrow 100 + 80 = 180$; $(0, 40) \rightarrow 160$. The maximum is at $(20, 20)$ with $Z = 180$. Option (d) identifies the correct point but the wrong value, so no option is fully correct and the answer is (e).

Part II — Python: Answer Key

Q37	Q38	Q39	Q40	Q41	Q42	Q43
(b)	(a)	(c)	(b)	(b)	(a)	(c)

Part II — Justifications

Q37. Answer: (b) `np.linspace(a, b, n)` returns n evenly spaced values from a to b **inclusive**. With $n = 5$ over $[0, 10]$ the spacing is $10/(5-1) = 2.5$. Option (c) describes `np.arange(0, 10, 2)`, which excludes the endpoint — a different function with a different convention.

Q38. Answer: (a) `sp.oo` is SymPy's symbol for infinity, so this computes $\lim_{t \rightarrow \infty} \frac{5000t}{t+20}$. Dividing numerator and denominator by t gives $\frac{5000}{1+20/t} \rightarrow 5000$. This is the ceiling the population approaches but never exceeds.

Q39. Answer: (c) `sp.Eq(demand, supply)` builds the equation $120 - 3P = 20 + 2P$, and `sp.solve(..., P)` solves it *for* P , returning a list of solutions: $[20]$. That is the equilibrium *price*. To get the equilibrium quantity you would substitute it back: $120 - 3(20) = 60$.

Q40. Answer: (b) `odeint` integrates the system numerically from the initial condition $[300, 15]$ across the supplied time array. It returns one row per time point and one column per state variable — here 2000×2 . It produces *numbers*, not a symbolic solution (that would be SymPy's job), and not the equilibrium.

Q41. Answer: (b) `comb(n, k)` is the binomial coefficient $\binom{n}{k}$, which counts *unordered* selections: $\binom{8}{3} = 56$. Option (a) is the permutation count $8 \times 7 \times 6 = 336$, which counts each committee $3! = 6$ times over.

Q42. Answer: (a) For $x(t) = A \cos(Bt) + D$: the amplitude is the multiplier $A = 3$, the vertical shift is $D = 7$, and the period is $T = 2\pi/B$. Here $B = \pi$, so $T = 2\pi/\pi = 2$ seconds. Option (d) inverts the relationship, giving the frequency rather than the period.

Q43. Answer: (c) `np.meshgrid` builds a grid of (x, y) coordinate pairs. Each comparison such as `X + Y <= 40` returns a boolean array of the same shape, and `&` combines them elementwise, so a point is `True` only where *both* constraints hold. Note this locates the feasible *region*; it does not evaluate the objective or find the optimum.

Part III — Answer Key (final answers only)

Candidates answer any **two** of these four; only the two attempted are marked. Listed below are the results you should arrive at — the working is yours to produce.

Q44. Lymphocyte count (12 marks)

- (a) $W(t) = -0.15t + 0.0455t^2 - 0.003t^3 + C$.
- (b) 2pm is $t = 6$, where $W(6) = 0.09 + C = 2.39$, so $C = 2.30$ and $W(t) = -0.15t + 0.0455t^2 - 0.003t^3 + 2.30$.
- (c) $W(0) = \mathbf{2.30}$ and $W(10) = \mathbf{2.35}$ (billion cells/L).
- (d) Total change $= \int_0^{10} W'(t) dt = W(10) - W(0) = +\mathbf{0.05}$. The count is marginally *higher* at 6pm than at 8am, but the change across the day is small relative to the level, so the time of day matters little for scheduling — though the count does dip and recover in between.

Q45. Deer fattening (12 marks)

- (a) $w(t) = 50 + 2t - 0.05t^2$ kg per animal; revenue $= 500w(t)$, cost $= 50t$, so $\pi(t) = 25,000 + 950t - 25t^2$.
- (b) $t^* = \mathbf{19}$ days; $\pi''(t) = -50 < 0$ confirms a maximum.
- (c) $w(19) = 69.95$ kg; total revenue $= \mathbf{\$34,975}$; total cost $= \mathbf{\$950}$; profit $= \mathbf{\$34,025}$.

Q46. Demand and supply (12 marks)

- (a) Choke price: $P = \mathbf{\$50}$.
- (b) Minimum supply price: $P = \mathbf{\$33.33}$.
- (c) $P^* = \mathbf{\$40}$, $Q^* = \mathbf{20}$ units.
- (d) $CS = \frac{1}{2}(50 - 40)(20) = \mathbf{\$100}$.
- (e) $PS = \frac{1}{2}(40 - 33.33)(20) = \mathbf{\$66.67}$.

Q47. Diet linear program (12 marks)

- (a) Minimise $Z = 2x + 3y$ subject to $4x + 5y \geq 100$ (protein), $10x + 25y \geq 400$ (energy), $0.8x + 0.4y \geq 10$ (vitamin A), $y \geq 5$ (calcium), $x, y \geq 0$.
- (b) Feasible vertices: $(0, 25)$, $(4.17, 16.67)$, $(10, 12)$ and $(27.5, 5)$. Intersections such as $(5.625, 13.75)$ fail the protein constraint and are infeasible.
- (c) Costs at those vertices: $\$75$, $\$58.33$, $\$56$, $\$70$. The minimum is $(x, y) = (10, 12)$ — 10 units of Food I and 12 of Food II — at $\mathbf{\$56}$. Protein $(4(10) + 5(12) = 100)$ and energy $(10(10) + 25(12) = 400)$ are *binding*; vitamin A $(12.8 > 10)$ and calcium $(12 > 5)$ are *slack*.
- (d) With Food I at $\$5$ the objective becomes $Z = 5x + 3y$. The vertices are unchanged, but their costs are now $\$75$, $\$70.83$, $\$86$ and $\$152.50$, so the optimum *shifts* to $(x, y) = (4.17, 16.67)$ at $\mathbf{\$70.83}$ — less of the now-dearer Food I and more of Food II.

— END OF ANSWER KEY —